Circuit family × initialisation × dataset

A progressive block-size study on DIV2K, TU-Berlin and QuickDraw

Generated 2026-05-25 · mean test PSNR over the test split

Summary. We train five quantum-circuit transform families — `rich` (complex $U(4)$ gates), `qft`, `tebd`, `entangled_qft`, `mera` — under a block-size curriculum, from two initialisations (identity-operator and random), on three datasets: DIV2K (natural images, 256^2), TU-Berlin (high-res sketches, 256^2), and QuickDraw (low-res drawings, 32^2). **The dataset sets the regime**: on DIV2K the block-size curve is flat and `rich` leads; on block-sparse TU-Berlin it falls (small blocks win) and classical block-DCT is near-lossless; on coarse QuickDraw the global QFT-family wins outright and block-DCT is weak. **A common rate- crossover holds**: classical block-DCT dominates the learned circuits at light compression but they overtake it as compression tightens (and on QuickDraw the learned circuits win at every rate). `rich` leads on DIV2K, the QFT-family on the two drawing sets; **identity init is never worse than random** — often dramatically better.

1 Setup

Each `*_progressive` sweep trains a circuit family under a **block-size curriculum**: at stage k the inner basis is a $2^k \times 2^k$ circuit replicated across the image by a `BlockedBasis` wrapper (the bare full circuit at $k = 8$). Stages are independent — no warm-start — and each trains for 1008 steps (generalized preset, 112 epochs) under an `MSELoss` that keeps the top-k coefficients by magnitude.

Families	`rich` (complex $U(4)$ 2-qubit gates), `qft`, `tebd`, `entangled_qft`, `mera`
Initialisations	**identity** — every gate dropped to its manifold identity (H $\to I_2$, controlled-phase $\to$ phase 0), so the circuit starts as the identity operator. **random** — `rich`: Haar $U(2)/U(4)$; `qft`: Haar $U(2)$ on Hadamards + random controlled-phase angles; `tebd`/`entangled_qft`/`mera`: native seeded random gates.
Stage range	`rich`/`qft`: $k = 1..8$; `tebd`/`entangled_qft`: $k = 2..8$; `mera`: $k \in \{2, 4, 8\}$ (m must be a power of 2). Tables show $k = 2..8$.
Datasets	**DIV2K-8q** — 500 natural images, centre-crop + LANCZOS to 256×256. **TU-Berlin** — 500 line-drawing sketches, $\approx 96.5\%$ white pixels, same preprocessing.
Metric	mean test PSNR @ keep-ratio ρ (fraction of coefficients retained); $\rho = 0.20$ is the headline rate.

2 DIV2K-8q: natural images

family	init	k=2 (4)	k=3 (8)	k=4 (16)	k=5 (32)	k=6 (64)	k=7 (128)	k=8 (256)
rich	identity	32.9	33.7	33.6	33.4	33.4	33.6	33.6
rich	random	32.9	33.4	33.5	32.9	33.2	32.9	33.4
qft	identity	32	32.3	32	31.7	31.7	31.7	31.7
qft	random	32	32.3	31.7	30.9	30.8	31.3	31.3
tebd	identity	32	32.3	32	31.7	31.7	31.7	31.7
tebd	random	32	32.3	31.7	30.9	30.8	30.9	30.9
ent-qft	identity	32	32.3	32	31.7	31.7	31.7	31.7
ent-qft	random	32	32.3	31.7	30.9	31.3	31.3	31.3
mera	identity	32	n/a	32	n/a	n/a	n/a	31.7
mera	random	32	n/a	31.7	n/a	n/a	n/a	30.9

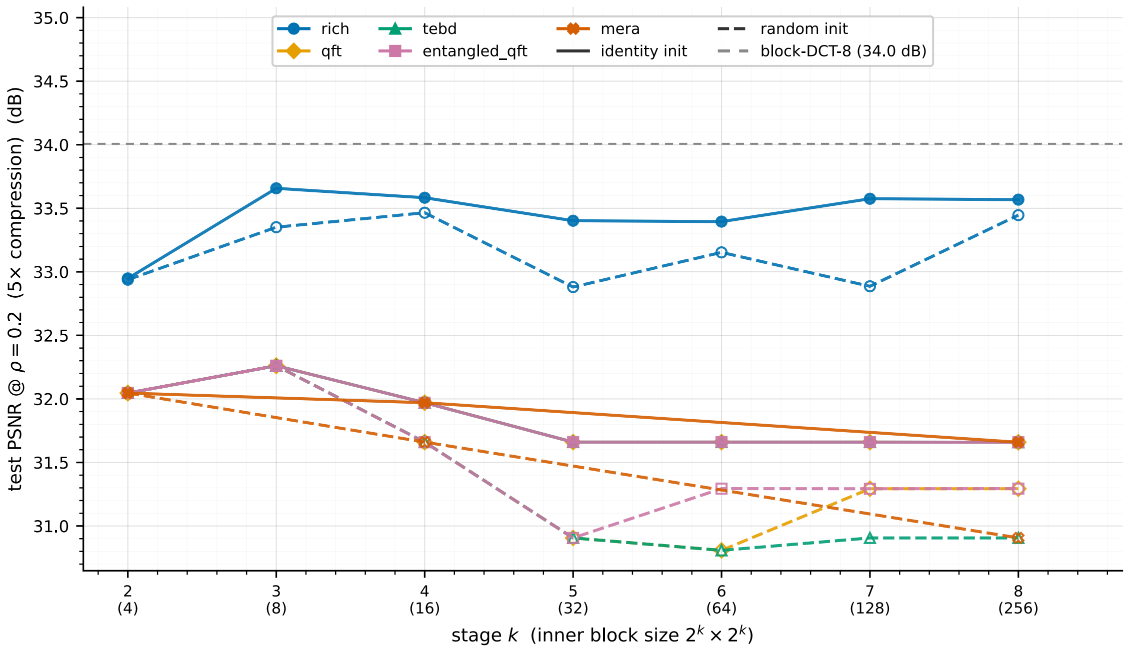

Figure 1: DIV2K, $\rho = 0.20$ (5×). One colour per family, solid = identity / dashed = random init; grey dashed = classical block-DCT-8 reference. `rich` leads the learned families but sits just below block-DCT.

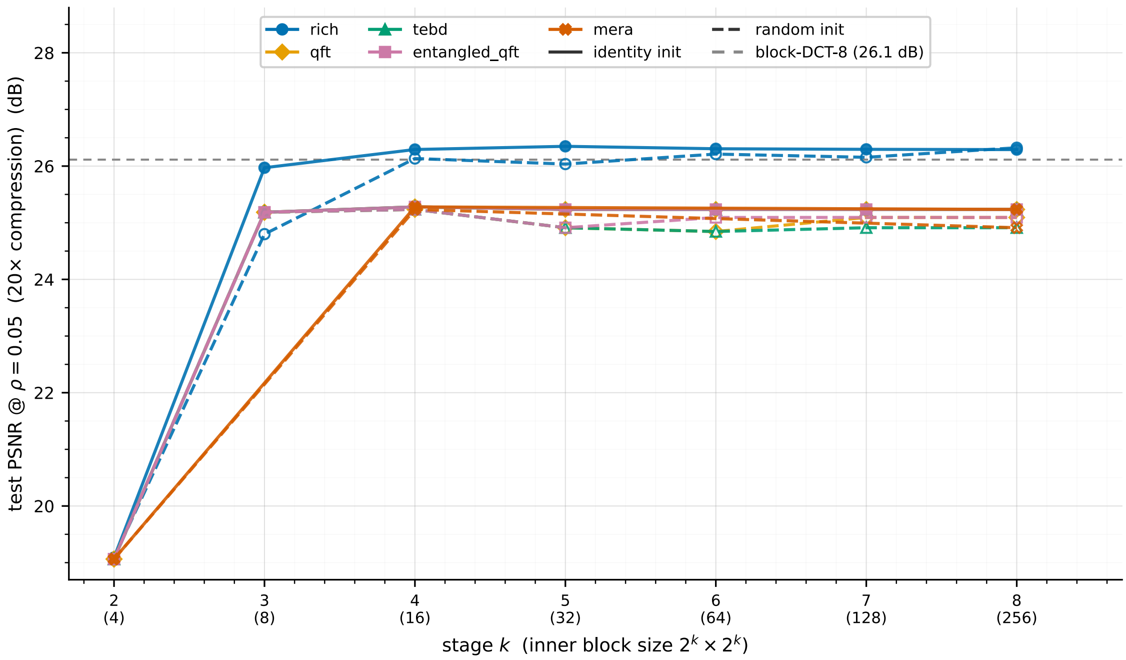

Figure 2: DIV2K, $\rho = 0.05$ (20×). The families bunch near block-DCT-8; `rich` draws level with it.

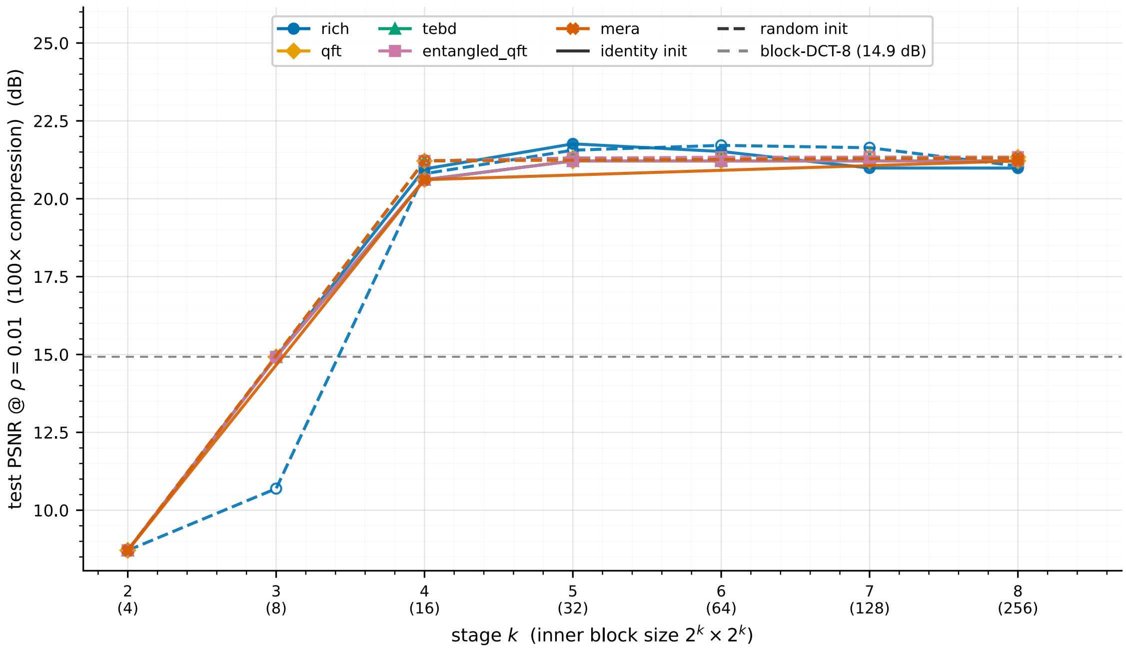

Figure 3: DIV2K, $\rho = 0.01$ (100×). Every learned family clears block-DCT-8 by $\approx$ 6–7 dB — the learned transforms win at aggressive compression.

Findings.

- **`rich` leads the learned families but classical block-DCT wins at the light rate.** At $\rho = 0.20$ `rich` peaks at 33.7 dB, $\approx$ 1.5 dB above every other family — yet still just below the classical block-DCT-8 reference (34.0 dB). Complex $U(4)$ is the one architectural feature that helps, but it does not beat the fixed block transform here.
- **The learned circuits overtake block-DCT as compression tightens.** block-DCT-8 falls faster with the rate (34.0 $\rightarrow$ 26.1 $\rightarrow$ 14.9 dB at $\rho = 0.20/0.05/0.01$) than the learned circuits do: `rich` draws level at $\rho = 0.05$ (26.4 vs 26.1) and **every** family clears block-DCT by $\approx$ 6–7 dB at $\rho = 0.01$ ($\approx$ 21 vs 14.9).
- **Under identity init the four QFT-derived families are bit-identical** at every k despite very different gate counts ($k = 8$: qft 72, tebd 32, entangled_qft 80, mera 44). From an identity start they all converge to the same QFT operator — the extra gates train to no-ops.
- **Identity init $\geq$ random** for every family; the curve is essentially flat in k ($\approx$ 33 dB for `rich`, $\approx$ 31.7 for the rest) at $\rho = 0.20$.

3 TU-Berlin: sketches (a different regime)

Sketches are **block-sparse** — most 8×8 tiles are blank — so a classical block transform is near-lossless: block-DCT-8 reaches 100.7 dB mean / 94.5 median at $\rho = 0.20$ (versus ≈ 32 on DIV2K), but only 36.5 dB at $\rho = 0.05$. We therefore report two compression rates.

3.1 Light compression: $\rho = 0.20$ (5×)

family	init	k=2 (4)	k=3 (8)	k=4 (16)	k=5 (32)	k=6 (64)	k=7 (128)	k=8 (256)
rich	identity	81.1	78.8	74.9	62.1	62.1	61.5	61.5
rich	random	79.5	73.8	62.6	48.8	41.5	33.9	44.9
qft	identity	83.6	87.8	71.4	51.6	50.1	64.2	62.6
qft	random	87.1	83.1	65.6	48.5	39.6	43.2	44.2
tebd	identity	85	87.5	69.6	51.4	50.2	62.9	62.9
tebd	random	88.2	87.2	69.5	45.9	34.6	44.2	38.1
ent-qft	identity	83.6	87.8	71.4	51.6	50.1	64.2	62.6
ent-qft	random	82.7	85.4	70.1	45.7	38.4	44.2	44.2
mera	identity	85	n/a	69.6	n/a	n/a	n/a	62.9
mera	random	88.2	n/a	69.4	n/a	n/a	n/a	38.4

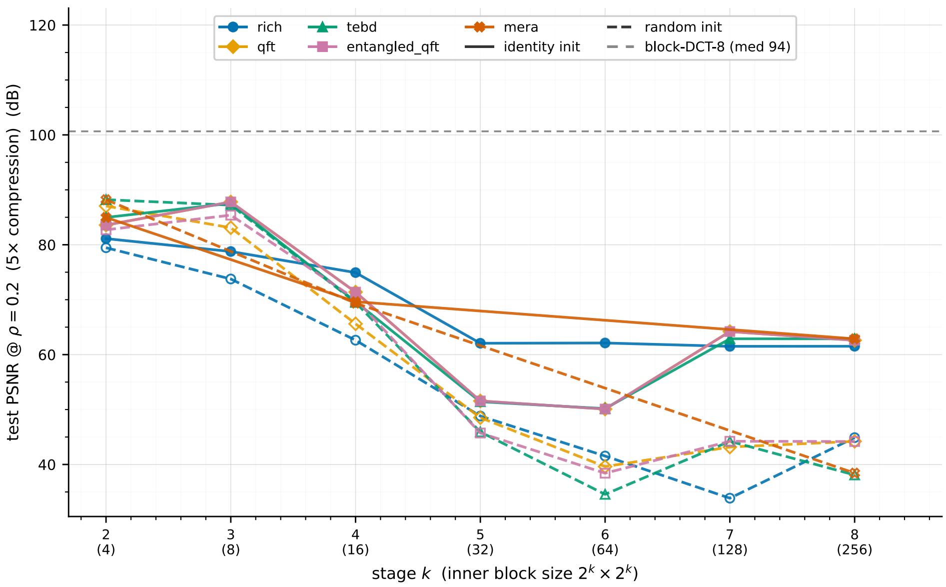

Figure 4: TU-Berlin, $\rho = 0.20$. The curve **falls** with block size; the QFT-derived families peak above `rich` at $k = 3$; all trail block-DCT-8 (grey).

3.2 Heavy compression: $\rho = 0.05$ (20×)

family	init	k=2 (4)	k=3 (8)	k=4 (16)	k=5 (32)	k=6 (64)	k=7 (128)	k=8 (256)
rich	identity	7.6	37.3	37.2	34.8	35.5	35.5	36.1
rich	random	7.6	30.6	31.6	29.6	28.5	25.7	29.3
qft	identity	7.6	33.6	31.5	29.7	29.7	31.5	31.5
qft	random	7.6	29.8	31	28.8	27.3	28	28.4
tebd	identity	7.6	33.5	31.5	29.7	29.7	31.5	31.5
tebd	random	7.6	33.5	31.5	28.5	26.1	28.5	27.1
ent-qft	identity	7.6	33.6	31.5	29.7	29.7	31.5	31.5
ent-qft	random	7.6	33.5	31.5	28.5	27.1	28.5	28.5
mera	identity	7.6	n/a	31.5	n/a	n/a	n/a	31.5
mera	random	7.6	n/a	31.5	n/a	n/a	n/a	27.1

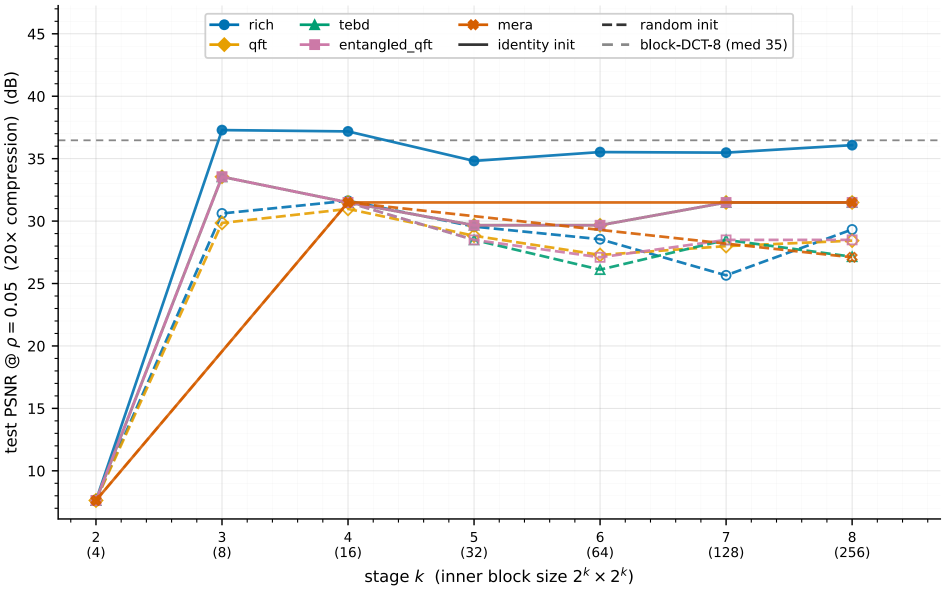

Figure 5: TU-Berlin, $\rho = 0.05$. `rich` (identity) leads and edges **above** block-DCT-8 at $k = 3, 4$; the per-k curve flattens.

3.3 Very heavy compression: $\rho = 0.01$ (100×)

family	init	k=2 (4)	k=3 (8)	k=4 (16)	k=5 (32)	k=6 (64)	k=7 (128)	k=8 (256)
rich	identity	0.9	4.8	4.8	22.1	21.7	22.2	22.9
rich	random	0.9	1.9	18.7	21.8	22.9	22.1	23
qft	identity	0.9	4.8	23.2	23.2	23.2	23.2	23.2
qft	random	0.9	1.9	21.8	22.9	22.5	22.5	23.1
tebd	identity	0.9	4.8	23.2	23.2	23.2	23.2	23.2
tebd	random	0.9	4.8	23.2	23.2	22.3	23.2	22.7
ent-qft	identity	0.9	4.8	23.2	23.2	23.2	23.2	23.2
ent-qft	random	0.9	4.8	23.2	23.2	22.7	23.2	23.2
mera	identity	0.9	n/a	23.2	n/a	n/a	n/a	23.2
mera	random	0.9	n/a	23.2	n/a	n/a	n/a	22.7

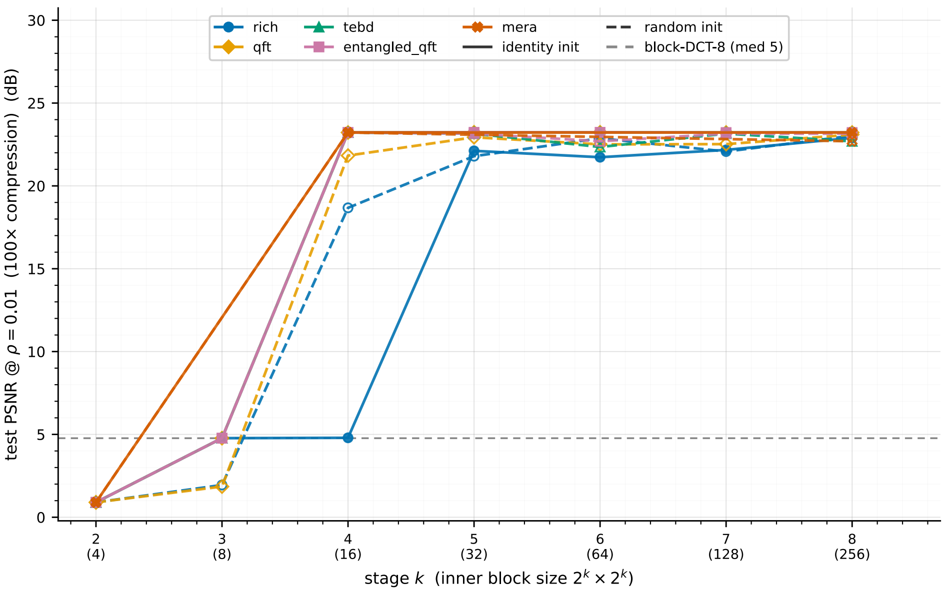

Figure 6: TU-Berlin, $\rho = 0.01$. block-DCT-8 **collapses** to ≈ 4.8 dB while the learned circuits plateau ≈ 23 dB — clearing it by ≈ 18 dB. An even sharper version of the heavy-compression crossover seen on DIV2K.

Findings.

- **The block-size curve inverts at the light rate.** Small blocks win — each blank tile costs almost no coefficients — the opposite of DIV2K's flat curve. (At the heavy rates the curves instead rise then plateau in k.)
- **The family ranking flips with the rate.** At $\rho = 0.20$ the QFT-derived families peak ≈ 88 dB at $k = 3$, above `rich` (≈ 79); at $\rho = 0.05$ `rich` (identity) leads (≈ 37 dB); at $\rho = 0.01$ the QFT-derived families edge back ahead (≈ 23.2 vs `rich` 22.9).
- **The heavy-compression crossover is extreme here.** block-DCT-8 falls $100.7 \to 36.5 \to 4.8$ dB across $\rho = 0.20/0.05/0.01$: it is near-lossless at the light rate but the learned circuits beat it by ≈ 1 dB at $\rho = 0.05$ and by ≈ 18 dB at $\rho = 0.01$.
- **Identity ⊳ random** — the gap reaches ≈ 25–30 dB at large k, far wider than on DIV2K. The QFT-family identity equivalence partly survives: `qft` $\equiv$ `entangled_qft` at all k, and all four QFT-derived families coincide at $\rho = 0.05$.

4 QuickDraw: low-resolution drawings

QuickDraw is 32×32 ($m = n = 5$), so the curriculum spans only $k = 1..5$ (block sizes 2..32; tables show $k = 2..5$, `mera` at $k \in \{2, 4\}$). The drawings are sparse strokes on a **dark** background (mean pixel ≈ 0.12), but at this low resolution they are far less block-compressible than the high-res TU-Berlin sketches: classical block-DCT-8 reaches only 26.63 dB at $\rho = 0.20$ (vs ≈ 100 on TU-Berlin).

4.1 Light compression: $\rho = 0.20$ (5×)

family	init	k=2 (4)	k=3 (8)	k=4 (16)	k=5 (32)
rich	identity	35.2	35.5	35.1	35
rich	random	34.2	32.8	32.8	29.3
qft	identity	39.8	39.6	39.4	39.4
qft	random	33.4	32.5	30.1	30.1
tebd	identity	39.8	39.6	39.4	39.4
tebd	random	30	30	30	24.4
ent-qft	identity	39.8	39.6	39.4	39.4
ent-qft	random	30	30.1	30.1	24.4
mera	identity	39.8	n/a	39.4	n/a
mera	random	30	n/a	30	n/a

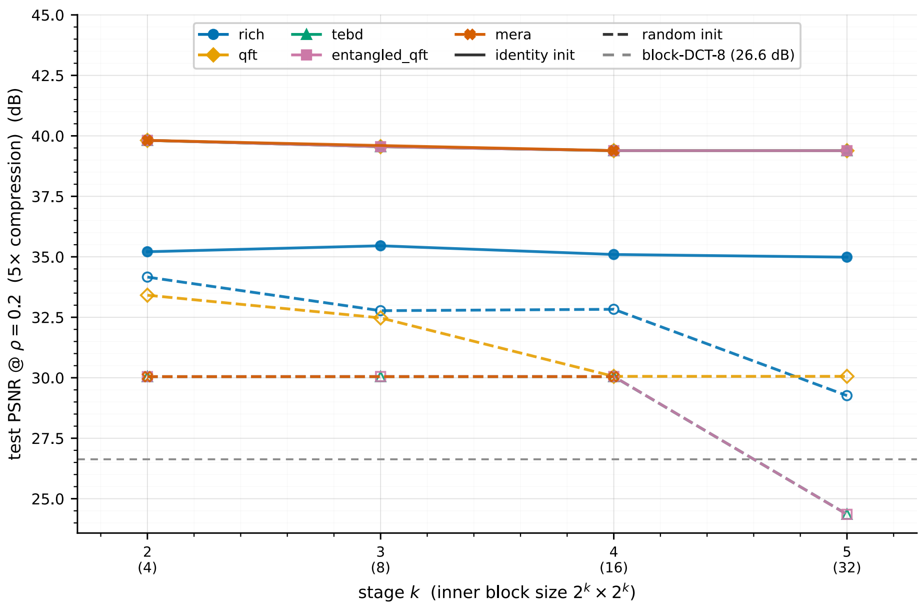

Figure 7: QuickDraw, $\rho = 0.20$. The QFT-derived families (identity, all four bit-identical at ≈ 39.5 dB) **dominate** — above `rich` (≈ 35) and well above block-DCT-8 (26.6). The curve is flat in k.

4.2 Heavy compression: $\rho = 0.05$ (20×)

family	init	k=2 (4)	k=3 (8)	k=4 (16)	k=5 (32)
rich	identity	17.8	17.8	17.8	17.8
rich	random	18.4	18.8	18.9	18.4
qft	identity	17.3	17.3	17.3	17.2
qft	random	17.8	17.3	18.1	18.1
tebd	identity	17.3	17.3	17.3	17.2
tebd	random	18.1	18.1	18.1	16.7
ent-qft	identity	17.3	17.3	17.3	17.2
ent-qft	random	18.1	18.1	18.1	16.7
mera	identity	17.3	n/a	17.3	n/a
mera	random	18.1	n/a	18.1	n/a

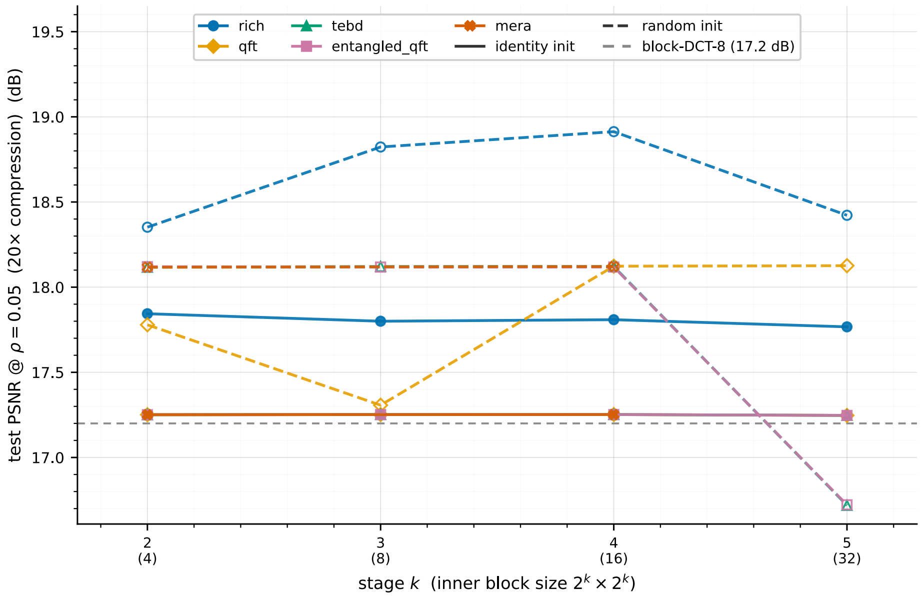

Figure 8: QuickDraw, $\rho = 0.05$. All families bunch near block-DCT-8 ($\approx$ 17 dB); the identity advantage vanishes and random init even edges ahead at some k.

4.3 Very heavy compression: $\rho = 0.01$ (100×)

family	init	k=2 (4)	k=3 (8)	k=4 (16)	k=5 (32)
rich	identity	11.5	11.5	11.5	11.5
rich	random	12.2	12.7	12.8	13.3
qft	identity	11.3	11.3	11.3	11.3
qft	random	12.1	11.9	12.6	12.6
tebd	identity	11.3	11.3	11.3	11.3
tebd	random	12.6	12.6	12.6	13
ent-qft	identity	11.3	11.3	11.3	11.3
ent-qft	random	12.6	12.6	12.6	13
mera	identity	11.3	n/a	11.3	n/a
mera	random	12.6	n/a	12.6	n/a

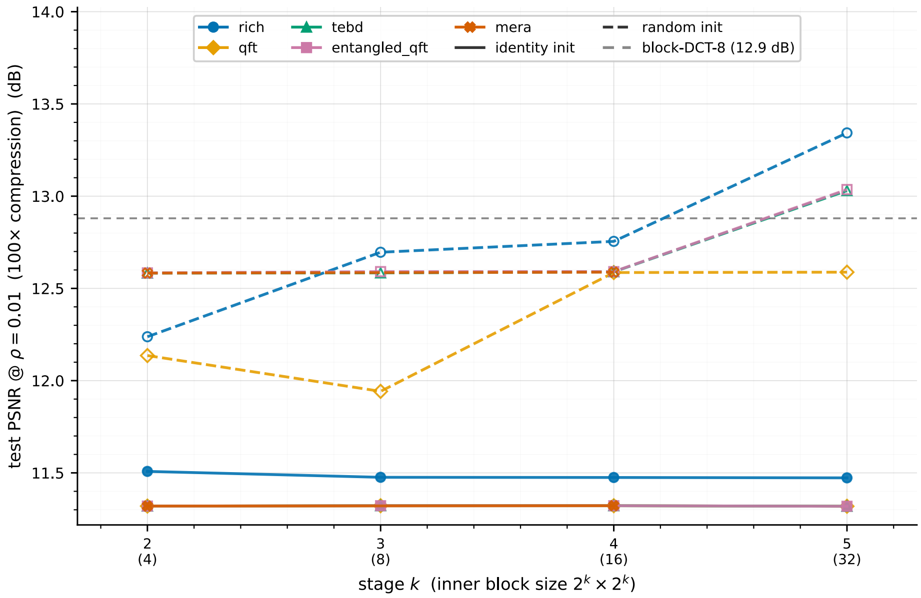

Figure 9: QuickDraw, $\rho = 0.01$. Everything collapses toward block-DCT-8 (≈ 12.9 dB); families are within ≈ 2 dB of each other.

Findings.

- **Learned circuits beat block-DCT at the light rate.** Unlike DIV2K/TU-Berlin, block-DCT-8 is weak on 32×32 drawings (26.6 dB), so at $\rho = 0.20$ the QFT-derived families (≈ 39.5) and even `rich` (≈ 35) clear it.
- **The QFT-derived families lead and are bit-identical under identity init,** beating `rich` by ≈ 4 dB at $\rho = 0.20$ — the global QFT solution suits these coarse drawings better than `rich`'s complex gates. The curve is flat in k (like DIV2K, unlike TU-Berlin's falling curve).
- **At heavy compression the families converge** to block-DCT (≈ 17 dB at $\rho = 0.05$, ≈ 12 at $\rho = 0.01$) and the identity advantage erodes — random init even edges ahead at some k.

5 Cross-dataset discussion

	DIV2K (natural, 256^2)	TU-Berlin (sketch, 256^2)	QuickDraw (drawing, 32^2)
block-size curve	flat in k	falls with k (small blocks win)	flat in k
best learned family	`rich`	QFT-family @ 0.2 & 0.01; `rich` @ 0.05	QFT-family
block-DCT @ ρ=0.2	34 dB (beats all)	94 dB (beats all)	27 dB (learned beat it)
vs block-DCT	learned overtake by ρ=0.01	learned overtake @ 0.05 & 0.01	learned win at all ρ @ 0.2; converge @ 0.01
identity vs random	identity $\geq$ random (<1 dB)	identity $\rhd$ random (25–30 dB)	identity wins @ 0.2; ties @ 0.05/0.01
PSNR @ ρ=0.2	$\approx$ 31–34	$\approx$ 60–88	$\approx$ 35–40

The contrast is governed by **energy compaction relative to the data's sparsity structure and resolution**. Natural-image energy is spread across scales, so a large circuit and the complex-$U(4)$ richness of `rich` help; high-res sketches are block-sparse, so a small per-block transform is near-lossless and the curve falls with k; low-res QuickDraw drawings are too coarse for 8×8 blocks to exploit, so the global QFT-family solution wins outright and block-DCT is weak. A common thread: the fixed block transform degrades faster than the learned circuits as the rate tightens, so learning pays off most at aggressive compression (and, on coarse QuickDraw, at every rate).

6 Conclusions

1. **The dataset sets the regime.** "Which circuit family is best" does not transfer across natural images, high-res sketches and low-res drawings; even the sign of the block-size trend and the leading family change.

2. **Learned circuits beat classical block-DCT at aggressive compression** (and at every rate on coarse QuickDraw, where block-DCT is weak); on the 256^2 sets block-DCT wins at the light rate but the learned transforms overtake it as ρ shrinks.
3. **Architectural richness is dataset-dependent:** complex $U(4)$ (`rich`) leads on natural images, but the plain QFT-family solution leads on both drawing sets.
4. **Identity initialisation dominates random** on the 256^2 sets (decisively on sketches); on coarse QuickDraw the advantage holds only at the light rate and ties under heavy compression. The structured identity start is a strong, cheap prior.

Data: `results/family_init_matrix/{div2k_8q,tuberlin_8q,quickdraw_5q}/<family>_<init>/`. Reproduce a sweep with `experiments/qft_progressive.py --dataset <d> --family <f> --init <i>`; regenerate figures with each report's `render_fig*.py`.